Josephson Parametric Amplifiers and Traveling-Wave Parametric Amplifiers

Quantum-Limited Microwave Amplification for Superconducting Qubit Readout

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Outline

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   - Amplifiers in Superconducting Qubit Readout
   - Phase-Sensitive vs. Phase-Insensitive Amplification
2. Josephson Parametric Amplifier
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Amplifiers in Superconducting Qubit Readout

Room-temperature microwave pulses are strongly attenuated before reaching the qubit in order to suppress thermal noise, thermalize the input line, and set the proper readout power. After interacting with the qubit-resonator system, the reflected or transmitted microwave signal is extremely weak. Therefore, cryogenic low-noise amplification is required to boost the signal before further room-temperature processing.

Therefore, qubit readout requires low-noise amplification. As quantum processors scale to larger numbers of qubits, the amplifier must also support multiplexed readout over a wider bandwidth. Because space and cooling power at the millikelvin stage are limited, broadband and high-dynamic-range amplifiers are highly desirable.

Several types of microwave amplifiers have been used in superconducting qubit readout chains, including cryogenic semiconductor amplifiers such as **HEMTs, resonator-based Josephson parametric amplifiers, and traveling-wave parametric amplifiers** based on Josephson junctions or kinetic inductance. Each technology offers a different tradeoff among gain, added noise, bandwidth, saturation power, and scalability.

$$\text{Qubit} \longrightarrow \text{JPA or TWPA} \longrightarrow \text{Circulator / Isolator} \longrightarrow \text{HEMT} \longrightarrow \text{Room-temperature electronics}$$

Interaction Hamiltonian of $\chi^{(2)}$ and $\chi^{(3)}$ non-linearity

We introduce these concepts starting from nonlinear optics. For a material with nonlinear optical response, the polarization can be written as

$$\mathbf{P} = \varepsilon_0 \left[\chi^{(1)}\mathbf{E} + \chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \cdots \right]. \tag{1}$$

Where we have the interaction Hamiltonian for the second- and third-order nonlinearity terms

$$H_{\chi^{(2)}} \sim \int \mathrm{d}^3 r \varepsilon_0 \chi^{(2)} E^3, \quad H_{\chi^{(3)}} \sim \int \mathrm{d}^3 r \varepsilon_0 \chi^{(3)} E^4. \tag{2}$$

After canonical quantization of the electric field, for pump, signal, and idler modes we write

$$\hat{\mathbf{E}}(\mathbf{r}, t) = A(\mathbf{r}) \sum_m \left(\hat{a}_m e^{-i\omega_m t} + \hat{a}_m^\dagger e^{i\omega_m t} \right), \quad m = p, s, i, \tag{3}$$

After RWA, we only keep slow rotating terms, the interaction Hamiltonians become

$$H_{\chi^{(2)}}/\hbar = g_2 \, \hat{a}_p \hat{a}_s^\dagger \hat{a}_i^\dagger + g_2^* \, \hat{a}_p^\dagger \hat{a}_s \hat{a}_i, \tag{4}$$

$$H_{\chi^{(3)}}/\hbar = g_3 \, \hat{a}_p^2 \hat{a}_s^\dagger \hat{a}_i^\dagger + g_3^* \, \hat{a}_p^{\dagger 2} \hat{a}_s \hat{a}_i, \tag{5}$$

with frequency conditions $\omega_p = \omega_s + \omega_i$ for three-wave mixing, and $2\omega_p = \omega_s + \omega_i$ for four-wave mixing.

Degenerate Parametric Amplification

Parametric Regime

Parametric regime refers to a situation where one or more modes of the system are driven strongly by a classical pump field. The corresponding quantum operators can be approximated by complex classical amplitudes.

Degenerate case

One pump photon at frequency ω_p splits into two signal photons at frequency ω_s, with

$$\omega_p = 2\omega_s.$$

Take three-wave mixing as an example. For a degenerate parametric amplifier, the full Hamiltonian can be written as (g is taken to be real for simplicity)

$$H = \hbar\omega_s \hat{a}_s^\dagger \hat{a}_s + \hbar\omega_p \hat{a}_p^\dagger \hat{a}_p + \hbar g \left(\hat{a}_s^{\dagger 2} \hat{a}_p + \hat{a}_s^2 \hat{a}_p^\dagger\right), \tag{6}$$

In the parametric regime, we take $\hat{a}_p \to \alpha_p e^{-i\omega_p t}$.

Moving into Rotating Frame and Phase Sensitive Amplification

Then the effective Hamiltonian for the signal mode becomes

$$H_{\text{eff}} = \hbar\omega_s \hat{a}_s^\dagger \hat{a}_s + \hbar\lambda \left(\hat{a}_s^{\dagger 2} e^{-i\omega_p t} + \hat{a}_s^2 e^{i\omega_p t}\right) \tag{7}$$

where $\lambda = g\alpha_p$. Then, in rotating frame, The interaction Hamiltonian can be written as

$$H_{\text{I}} = \hbar\lambda \left(\hat{a}_s^{\dagger 2} + \hat{a}_s^2\right). \tag{8}$$

which is a squeezing Hamiltonian.

Phase Sensitive Amplification

By analyzing the dynamics of field operator $\hat{a}_s$, then quadratures of this mode X_1 X_2 via Heisenberg Equation. We can see the quadratures is squeezed and anti-squeezed.

$$X_1(t) = e^{\lambda t} X_1(0) \tag{9}$$

$$X_2(t) = e^{-\lambda t} X_2(0) \tag{10}$$

These equations demonstrate the degenerate parametric down-conversion process is a parametric amplifier which is phase-sensitive.

Phase-Sensitive Amplification

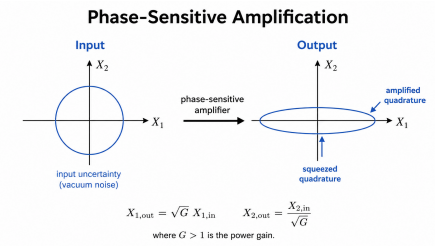
Next, we analyze the noise property of phase-sensitive amplification.

Input-Referred Noise

The input vacuum variance is $\mathrm{Var}(X_i) = 1/4$ (for $i = 1, 2$).
The output quadrature variance, referred back to input, is:

$$N_{\mathrm{add}} = \frac{\mathrm{Var}(X_1^{\mathrm{out}})}{G} - \mathrm{Var}(X_1^{\mathrm{in}}) = 0, \tag{11}$$

demonstrating noiseless amplification along the amplified quadrature.



One quadrature is amplified, while the orthogonal quadrature is squeezed.

SNR Enhancement for Amplified Quadrature

Since only the amplified quadrature carries signal, the signal-to-noise ratio improves proportionally to the gain G:

$$\mathrm{SNR}_{\mathrm{out}} = G\mathrm{SNR}_{\mathrm{in}}. \tag{12}$$

Non-Degenerate Parametric Amplification

Non-Degenerate Case

Non-degenerate parametric down-conversion involves three different modes: signal, idler, and pump. The pump mode at frequency ω_p is converted into one signal photon at ω_s and one idler photon at ω_i, satisfying the frequency-matching condition

$$\omega_p = \omega_s + \omega_i.$$

In contrast to the degenerate case, the signal and idler modes have different frequencies.

Recall the full three-wave mixing Hamiltonian ($\chi^{(2)}$):

$$H = \hbar\omega_1 \hat{a}_1^\dagger \hat{a}_1 + \hbar\omega_2 \hat{a}_2^\dagger \hat{a}_2 + \hbar\omega_3 \hat{a}_3^\dagger \hat{a}_3 + \hbar g(\hat{a}_1^\dagger \hat{a}_2^\dagger \hat{a}_3 + \hat{a}_1 \hat{a}_2 \hat{a}_3^\dagger), \tag{13}$$

where $\omega_3 = \omega_1 + \omega_2$. In the parametric regime, the effective interaction between signal and idler:

$$H_{\text{eff}} = \hbar\omega_1 \hat{a}_1^\dagger \hat{a}_1 + \hbar\omega_2 \hat{a}_2^\dagger \hat{a}_2 + \hbar\lambda(\hat{a}_1^\dagger \hat{a}_2^\dagger e^{-i\omega_3 t} + \hat{a}_1 \hat{a}_2 e^{i\omega_3 t}). \tag{14}$$

This Hamiltonian describes non-degenerate parametric down-conversion, where one pump photon at ω_3 creates a signal photon at ω_1 and an idler photon at ω_2.

Moving into Rotating Frame and Two Mode Squeezing

In rotating frame, the interaction Hamiltonian can be written as

$$H_{\mathrm{I}} = \hbar\lambda \left(\hat{a}_1^{\dagger} \hat{a}_2^{\dagger} + \hat{a}_1 \hat{a}_2 \right) \tag{15}$$

Two Mode Squeezing

By analyzing the dynamics of field operator $\hat{a}_s$, then quadratures $X_i^{\theta} = \hat{a}_i e^{i\theta} + \hat{a}_i^{\dagger} e^{-i\theta}$ via Heisenberg Equation. Then define two mode quadrature $X^{\theta} = X_1^{\theta} + X_2^{\theta}$, we can see the sum of quadratures of the two modes is squeezed or anti-squeezed, naming two-mode squeezing.

$$X^0(t) = e^{\lambda t} X^0(0) \tag{16}$$

$$X^{\frac{\pi}{2}}(t) = e^{-\lambda t} X^{\frac{\pi}{2}}(0) \tag{17}$$

Phase Insensitive Amplification

Added Noise

For a phase-insensitive amplifier,

$$\hat{a}_{s,\mathrm{out}} = \sqrt{G}\,\hat{a}_{s,\mathrm{in}} + \sqrt{G-1}\,\hat{a}^{\dagger}_{i,\mathrm{in}} \tag{18}$$

$$X_{s,\mathrm{out}} = \sqrt{G}\,X_{s,\mathrm{in}} + \sqrt{G-1}\,X_{i,\mathrm{in}} \tag{19}$$

If the idler is in vacuum, $\mathrm{Var}(X_{i,\mathrm{in}}) = \frac{1}{4}$.

$$\mathrm{Var}(X_{s,\mathrm{out}}) = G\,\mathrm{Var}(X_{s,\mathrm{in}}) + (G-1)\frac{1}{4}. \tag{20}$$

The input-referred added noise for one quadrature is

$$A_X = \frac{\mathrm{Var}(X_{s,\mathrm{out}})}{G} - \mathrm{Var}(X_{s,\mathrm{in}}) = \frac{G-1}{G}\frac{1}{4} \xrightarrow[G\gg 1]{} \frac{1}{4}. \tag{21}$$

In photon-number units, the minimum added noise is

$$N_{\mathrm{add}} = 2A_X = \frac{G-1}{G}\frac{1}{2} \xrightarrow[G\gg 1]{} \frac{1}{2}. \tag{22}$$

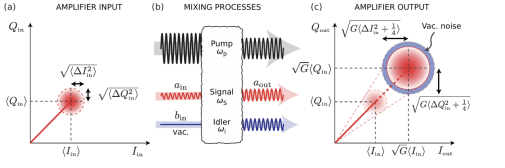


FIG. 25. Schematic illustration of a quantum-limited, phase-preserving parametric amplification process of a coherent input state, $a_{\mathrm{in}} = I_{\mathrm{in}} + iQ_{\mathrm{in}}$. **(a)** The state is centered at $(\langle I_{\mathrm{in}}\rangle, \langle Q_{\mathrm{in}}\rangle)$ and has a noise represented by the radii of the circles along the real and imaginary axes, respectively. **(b)** Scattering representation of parametric mixing, where the signal and pump photons are interacting via a purely dispersive nonlinear medium. **(c)** In the case of phase-preserving amplification, both quadratures get amplified by a factor $\sqrt{G}$, while (in the ideal case) half a photon of noise gets added to the output distribution (blue). Image inspired by Flurin[39].

Figure 1: Phase-preserving amplification adds idler vacuum noise. Adapted from Krantz et al.[1]

Comparison between Phase-Sensitive and Phase-Insensitive Amplification

Phase-Sensitive Amplification

- Degenerate case: $\omega_p = 2\omega_s$.
- Signal and idler are the same mode.
- One quadrature is amplified; the other is squeezed.
- Gain depends on input phase relative to pump.
- Ideal added noise: $N_{\text{add}} = 0$ for the amplified quadrature.

Phase-Insensitive Amplification

- Non-degenerate case: $\omega_p = \omega_s + \omega_i$.
- Signal and idler are different modes.
- Both quadratures are amplified equally.
- Gain is independent of input phase.
- Quantum-limited added noise $N_{\text{add}} \geq \frac{1}{2}$.

Key Difference

Phase-sensitive amplification can be noiseless for one quadrature, while phase-insensitive amplification preserves arbitrary signal phase but must add at least half a photon of noise. In the context of qubit readout, however, phase sensitive amplification tends to be experimentally in convenient. This is mainly due to its phase-dependent gain, which imposes stringent requirements on continous phase-calibration of the readout signal.[1]

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Overview of Different Implementation of JPA

- **Two main ways to apply the pump:**
  - Flux-pumped: modulates the SQUID inductance via an external magnetic flux.
  - Current- or voltage-pumped: applies a strong classical drive directly to the cavity or resonator.
- **Most commonly used:**
  - Reflected flux pumped.

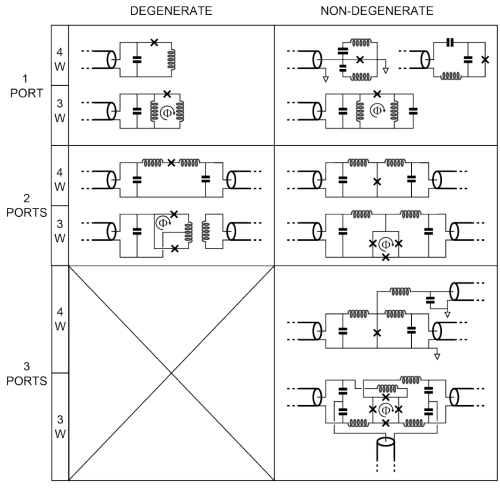


Figure 2: Different Implementation of JPA [2]

Two Ways of Pumping

As introduced in the previous section, parametric amplification requires a strong classical pump. The pump converts the Josephson nonlinearity into an effective mixing interaction. In Josephson parametric amplifiers, there are two common ways to introduce the pump.

Microwave Drive / Current Pump

A strong microwave tone is applied to the nonlinear resonator or mixer:

$$a_{\mathrm{in}}(t) = \alpha_p e^{-i\omega_p t} + \delta a_{\mathrm{in}}(t).$$

where we have the classical field $\alpha_p e^{-i\omega_p t}$ and readout signal $\delta a_{\mathrm{in}}(t)$.

Flux Pump / SQUID Inductance Modulation

A time-dependent magnetic flux is applied through a SQUID loop:

$$\Phi_{\mathrm{ext}}(t) = \Phi_{\mathrm{dc}} + \Phi_{\mathrm{ac}} \cos(\omega_p t).$$

This modulates the SQUID Josephson inductance and therefore the resonator frequency $L_J(t) \to \omega_r(t)$. The time-dependent circuit parameter acts as a parametric drive.

Microwave Drive / Current Pump

We demonstrate how a microwave drive induces an effective squeezing interaction using the simplest Duffing oscillator:

$$H/\hbar = \omega_a \hat{a}^\dagger \hat{a} + \frac{K}{2} \hat{a}^{\dagger 2} \hat{a}^2. \tag{23}$$

Under a strong microwave pump, the intracavity field can be separated into a large classical pump amplitude and a small quantum fluctuation:

$$\hat{a}(t) = \alpha e^{-i\omega_p t} + \delta\hat{a}(t), \tag{24}$$

Substituting this into the Kerr term and keeping only terms up to second order in $(\delta\hat{a})$, we obtain

$$\hat{a}^{\dagger 2}\hat{a}^2 \approx 4|\alpha|^2 \delta\hat{a}^\dagger \delta\hat{a} + \alpha^2 e^{-2i\omega_p t} \delta\hat{a}^{\dagger 2} + \alpha^{*2} e^{2i\omega_p t} \delta\hat{a}^2 + \text{const.} \tag{25}$$

Therefore, the effective Hamiltonian for the fluctuation is

$$H_{\text{eff}}/\hbar = \tilde{\omega}_a \delta\hat{a}^\dagger \delta\hat{a} + \frac{1}{2}\left[\lambda e^{-2i\omega_p t} \delta\hat{a}^{\dagger 2} + \lambda^* e^{2i\omega_p t} \delta\hat{a}^2\right], \tag{26}$$

where

$$\lambda = K\alpha^2. \tag{27}$$

Therefore, a strong microwave pump converts the original Kerr nonlinearity into an effective squeezing interaction for the weak signal fluctuation.

Flux-Pumped JPA: SQUID-Inductance Tuning

SQUID as a tunable inductance

A dc-SQUID can be treated as a flux-tunable Josephson inductance:

$$L_J(\Phi_{\mathrm{ext}}) = \frac{\Phi_0}{2\pi I_c \cos(\pi \Phi_{\mathrm{ext}}/\Phi_0)}. \tag{28}$$

Here Φ_{ext} is the external flux threading the SQUID loop, and $\Phi_0 = h/2e$.

Flux pump

Apply a time-dependent flux:

$$\Phi_{\mathrm{ext}}(t) = \Phi_{\mathrm{dc}} + \Phi_{\mathrm{ac}} \cos(\omega_p t). \tag{29}$$

Then the SQUID inductance and resonator frequency are modulated:

$$L_J(t) \simeq L_0 + \delta L \cos(\omega_p t), \tag{30}$$

$$\omega_r(t) \simeq \omega_a + \delta\omega \cos(\omega_p t). \tag{31}$$

Flux Pump as a Parametric Drive

For a resonator with capacitance C, the quadratic Hamiltonian is

$$H(t) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L_J(t)}. \tag{32}$$

Since the flux operator satisfies

$$\Phi \propto a + a^\dagger,$$

the time-dependent part of the Hamiltonian has the form

$$\delta H(t) \propto \cos(\omega_p t)(a + a^\dagger)^2. \tag{33}$$

Expanding the square,

$$(a + a^\dagger)^2 = a^2 + a^{\dagger 2} + 2a^\dagger a + 1. \tag{34}$$

When $\omega_p \approx 2\omega_a$, the rotating-wave approximation keeps the slowly rotating pair-creation terms:

$$H_{\text{eff}}/\hbar = \lambda a^{\dagger 2} e^{-i\omega_p t} + \lambda^* a^2 e^{i\omega_p t}. \tag{35}$$

Therefore, a time-dependent inductance directly produces a parametric interaction, leading to single-mode squeezing in the degenerate case.

Degenerate and Non-Degenerate Parametric Amplification

Different JPA designs have different circuit Hamiltonians, but the target is usually one of two effective forms:

Degenerate

Signal and idler occupy the same mode:

$$H_{\text{eff}}/\hbar = \lambda a^{\dagger 2} + \lambda^* a^2.$$

This gives phase-sensitive amplification.

Non-degenerate

Signal and idler occupy different modes:

$$H_{\text{eff}}/\hbar = \lambda a^\dagger b^\dagger + \lambda^* ab.$$

This can give phase-insensitive amplification.

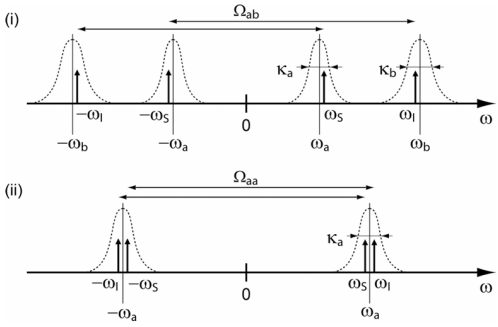


FIG. 1. Frequency landscape for the nondegenerate (i) and degenerate (ii) parametric amplifiers. The dashed lines correspond to response curves of each mode, as measured by a probe tone injected in the circuit elements of the mode. The vertical arrows correspond to the spectral densities of the signal and the idler tones arriving in the circuit through its ports. The horizontal arrows denote the frequency translations between signal and idler operated by the parametrical modulation induced by the pump tone.

Typical Performance of Josephson Parametric Amplifiers

Reference / Device	Type	Frequency	Gain	Bandwidth	Noise / Saturation
Yamamoto et al. 2008 [3]	Flux-driven SQUID JPA	$\sim 10\,\mathrm{GHz}$ signal	–	–	Flux pump at $\sim 20\,\mathrm{GHz}$
Bergeal et al. 2010 [4]	JRM / JPC, phase-preserving	microwave range	–	–	Within factor of 3 of quantum limit
Planat et al. 2019 [5]	80-SQUID-array JPA	$5.9{-}6.8\,\mathrm{GHz}$	$20\,\mathrm{dB}$	$45\,\mathrm{MHz}$	$P_{\mathrm{sat}} \approx -117\,\mathrm{dBm}$ input
Elo et al. 2019 [6]	Broadband lumped-element flux-pumped JPA	$\sim 5\,\mathrm{GHz}$, $> 1\,\mathrm{GHz}$ tunable	$20\,\mathrm{dB}$	$95\,\mathrm{MHz}$	Single-step lithography
Kaufman et al. 2023 [7]	Chebyshev impedance-matched JPA	$4.6\,\mathrm{GHz}$	$20\,\mathrm{dB}$	up to $500\,\mathrm{MHz}$	$P_{\mathrm{out,sat}} \approx -73\,\mathrm{dBm}$
Quantum Microwave / Raytheon BBN [8]	Commercial wideband JPA	$5.0{-}7.0\,\mathrm{GHz}$ tunable	$\geq 20\,\mathrm{dB}$	$300\,\mathrm{MHz}$	$T_N = 295\,\mathrm{mK}$ at $6.8\,\mathrm{GHz}$

Table 1: Selected representative JPA performance values from literature and commercial specifications.

What Do These Typical Values Tell Us?

High gain and near-quantum-limited noise

JPAs are very strong first-stage cryogenic amplifiers because they can provide high gain while adding very little noise. Typical operation is around $G \sim 20\,\mathrm{dB}$ which corresponds to a power gain of about 100. This is large enough to suppress the noise contribution from later-stage amplifiers such as the HEMT [9]:

$$N_{\mathrm{sys}} \approx N_{\mathrm{JPA}} + \frac{N_{\mathrm{HEMT}}}{G_{\mathrm{JPA}}}.$$

This is the main reason JPAs are widely used in superconducting qubit readout chains.

Bandwidth is improved, but still architecture-dependent

Traditional resonator-based JPAs often have bandwidths in the range of tens of MHz. Modern impedance-engineered designs can push the bandwidth to hundreds of MHz, for example 300–$500\,\mathrm{MHz}$. This shows that JPA bandwidth can be improved by engineering the impedance environment and gain profile. However, the amplifier is still based on resonant or matching-network structures. Therefore, its usable bandwidth is still limited compared with broadband traveling-wave amplification.

System-Level Limitation of JPA-Based Readout

Saturation power matters for multiplexed readout (Dynamic Range)

For a single readout tone, the input power can be kept low. For multiplexed readout, many tones enter the amplifier simultaneously:

$$P_{\mathrm{tot}} \approx \sum_i P_i.$$

If P_{tot} approaches the saturation power of the JPA, gain compression and distortion can occur, reducing the readout SNR.

Frequency allocation constraint

In a multiplexed readout line, several readout resonators share the same output chain: $f_1,\ f_2,\ \ldots,\ f_N$. If a JPA is used as the first-stage amplifier, these readout tones should ideally fall inside the usable JPA gain band:

$$f_1, \ldots, f_N \in B_{\mathrm{JPA}}.$$

Otherwise, different qubits may experience different amplification gain and different system noise. This creates constraints on how readout resonator frequencies can be assigned on the chip.

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Josephson Traveling-Wave Parametric Amplifier: Basic Principle

Distributed Non-linearity

Instead of a strong nonlinearity in JPA. TWPA distribute the nonlinearity over the transmission line. Each section on the transmission line provides only weak nonlinear gain, then the signal can be gradually amplified as it propagates along the device.

Phase Matching

On the TWPA, Pump, signal, and idler waves propagate with different wavevectors k_p, k_s, and k_i. n four-wave mixing, the source term driving the signal is $A_p^2 A_i^* e^{i(2k_p - k_i)z}$. To add coherently to the existing signal wave, this source must have the same spatial phase as $A_s e^{ik_s z}$. Therefore,

$$\Delta k = 2k_p - k_s - k_i \approx 0.$$

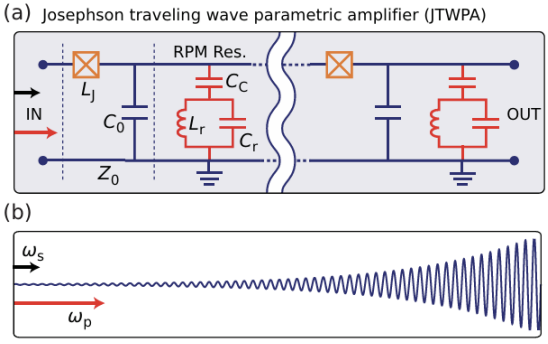


Figure 4: (a)Simplified circuit representation of a Josephson traveling wave parametric amplifier(JTWPA). (b) Schematic of how the signal gets amplified in each unit cell as it propagates through the device. [1]

Phase Matching

For transmission line, we have

$$\frac{\partial V}{\partial x} = -L' \frac{\partial I}{\partial t}, \quad \frac{\partial I}{\partial x} = -C' \frac{\partial V}{\partial t} \tag{36}$$

where L' and C' are inductance and capacitance per unit length, we can obtain the wave function

$$\frac{\partial^2 V}{\partial x^2} = L' C' \frac{\partial^2 V}{\partial t^2} \tag{37}$$

Let solution be $V(x,t) = V_0 e^{i(kx-\omega t)}$, plug in, we obtain

$$k = \omega \sqrt{L' C'} \tag{38}$$

we also have characteristic impedance given by

$$Z_0 = \sqrt{\frac{L'}{C'}} \tag{39}$$

Thus, by changing L', C', we can change both k and Z_0, and for k, we can therefore change the phase matching in the transmission line by

$$\phi(\omega) = \int k(x,\omega)\, \mathrm{d}x. \tag{40}$$

Resonator Phase-Matched JTWPA I

As demonstrated theoretically, when a traveling wave propagates through a non-linear transmission line with a Josephson junction array, it is possible to interact via a four-wave mixing process [10]. To address the phase matching challenge, Kevin O' Brien et al. 2014 [11] introduced a resonator technique.

Resonator Phase Matching

Each loaded LC resonator produces a strong dispersion feature near its resonance (Fig. 5). As a result, the dispersion curve $k(\omega)$ is bent near the pump frequency, giving the pump an additional phase shift (Fig. 6). This extra phase shift compensates the nonlinear phase mismatch: $\Delta k = k_s + k_i - 2k_p + \Delta k_{\mathrm{NL}} \to 0$. When Δk is closer to zero, the signal and idler generated at different positions add more coherently.

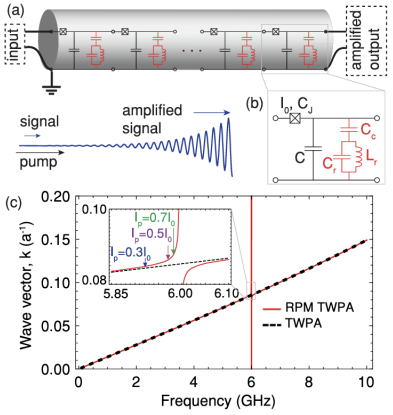


Figure 5: [11]

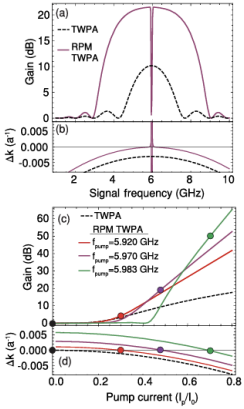


Figure 6: [11]

Resonator Phase-Matched JTWPA: Performance and Limitations

One of the most famous demonstrations was performed by Macklin *et al.* [12], based on the resonant phase-matching proposal by O'Brien *et al.* [11]. By periodically loading resonators into the Josephson transmission line, the dispersion near the pump frequency is engineered to satisfy the phase-matching condition.

Main advantage over resonator JPAs

A representative SQUID-array JPA has

$$G \approx 20\,\mathrm{dB}, \qquad B \approx 45\,\mathrm{MHz}, \qquad P_{\mathrm{sat,in}} \approx -117\,\mathrm{dBm}.$$

Resonantly phase-matched JTWPAs can reach

$$G \approx 20\,\mathrm{dB}, \qquad B \approx 3\,\mathrm{GHz}, \qquad P_{\mathrm{sat}} \approx -98\,\mathrm{dBm}.$$

This corresponds to roughly $60\times$ larger bandwidth and about $20\,\mathrm{dB}$ higher power handling.

Remaining limitations

- **Gain ripple from reflections and impedance mismatch.** Resonator loading introduces strong dispersion features and stop bands for phase matching, but it also makes the gain profile sensitive to impedance mismatch, producing ripples in $G(\omega)$ [13, 14].
- **Finite saturation power.**[5, 11].
- **Intermodulation distortion in multiplexed readout.** The nonlinear transmission line can mix readout tones and generate unwanted intermodulation products. This effect becomes more significant near saturation and can cause crosstalk and reduced readout fidelity [15].
- **Sensitivity to fabrication disorder.**

CP-JTWPA [13] I

Another implementation of JTWPA performs phase matching through impedance modulation. As discussed before, the distributed circuit parameters L' and C' determine both the characteristic impedance and the wave vector. Therefore, engineering L' and C' along the transmission line can shape both impedance matching and phase matching.
Chang *et al.* [13] implemented this idea using a low-loss coplanar lumped-element waveguide. The unit-cell parameters are controlled by the coplanar/open-stub capacitors and Josephson junction layout.

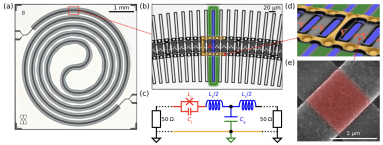


Figure 7: [13]

Reported performance

With Tukey-windowed modulation and 8% impedance variation, the ideal matched design reaches 20–$23\,\mathrm{dB}$ gain over a $5\,\mathrm{GHz}$ bandwidth. With practical impedance mismatches, the device still maintains 17–$20\,\mathrm{dB}$ gain over $4.8\,\mathrm{GHz}$ bandwidth.

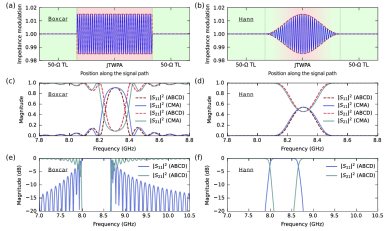


Figure 8: [13]

CP-JTWPA [13] II

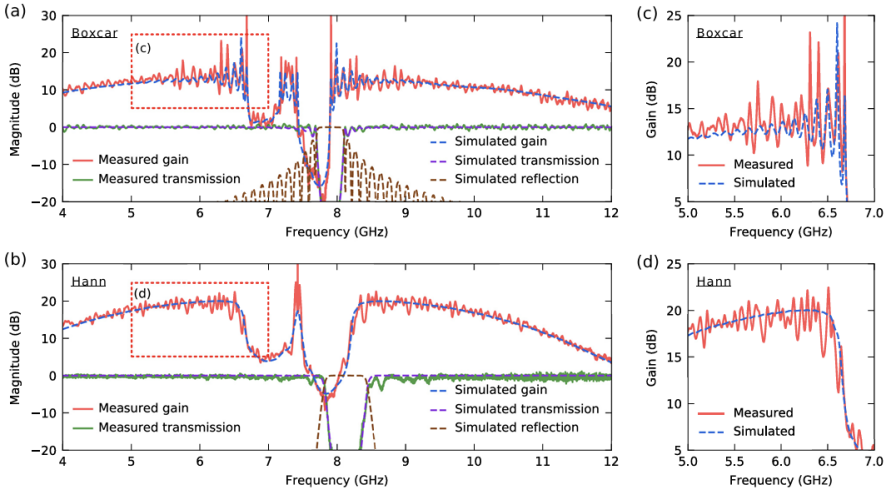


Figure 9: [13]

CP-JTWPA [13] III

Although Hann window can reduce edge reflections caused by the modulation (thereby reducing gain ripple), it also decreases the modulation amplitude in the central region, leading to a lower peak gain. In addition, the steep transitions at the edges of the bandwidth can introduce local non-ideal phase variations. In the article, the authors propose using a Tukey window (a flat-top window with gradual transitions) to replace the Hann window. The reasons are: Maintain maximum modulation in the central region, ensuring that the peak gain does not decrease. Smooth transitions at the edges, reducing reflections and discontinuities. Provide a compromise design: the flat-top center with gradual edge transitions reduces gain ripple while preserving broadband gain.

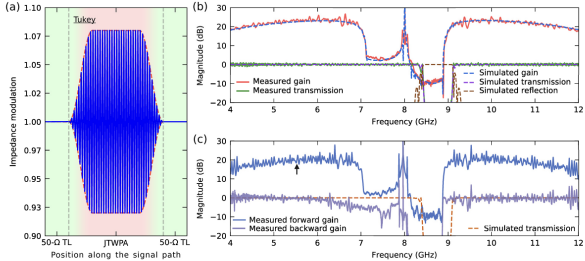


FIG. 4. Characterization of device C implementing a windowed periodic impedance modulation with a Tukey window with equal length in taper and maximum-amplitude region ($\alpha = 0.5$). (a) The impedance modulation profile along the signal path. The modulation amplitude increases smoothly from zero at the input to 8% in the central uniform section, then decreases symmetrically to zero at the output. (b) Measured and simulated scattering parameters with 10-dB attenuators at both ports providing 50-Ω impedance matching. (c) Measured forward and backward scattering parameters without output attenuation (see Appendix G for the corresponding microwave network), representing a more realistic microwave environment with typical circuit impedance mismatches present in applications. The noise measurement was performed at 5.563 GHz (arrow) under this configuration.

Figure 10: [13]

SNAIL based TWPA[17] I

Another approach is based on the Superconducting Nonlinear Asymmetric Inductive eLement (SNAIL), proposed by Frattini *et al.* [16]. A SNAIL is an asymmetric Josephson loop, similar in circuit layout to a flux qubit, but with parameters chosen to create a single-well potential. This design enhances the cubic nonlinearity while suppressing the unwanted Kerr term. Thus, in the parametric regime, the SNAIL naturally enables three-wave mixing.

The potential of a SNAIL can be expanded as

$$U(\varphi) = c_2\varphi^2 + c_3\varphi^3 + c_4\varphi^4 + \dots$$

With proper flux bias and junction asymmetry, c_3 is large while c_4 is reduced, enabling 3-wave mixing with minimal Kerr nonlinearity.

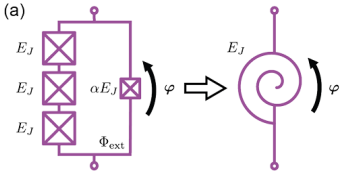


Figure 11: A SNAIL[16]

SNAIL based TWPA[17] II

Fadavi *et al.* and Renberg *et al.* proposed a SNAIL-based TWPA design in Ref. [17, 18]. Compared with single-pump four-wave mixing, an important advantage of three-wave mixing is that the pump can be well separated from the signal and idler band. For three-wave mixing,

$$\omega_p = \omega_s + \omega_i,$$

while for single-pump four-wave mixing,

$$2\omega_p = \omega_s + \omega_i.$$

Thus, in 4WM the pump lies near the middle of the signal-idler band, making pump leakage and filtering more challenging. In contrast, 3WM places the pump far from the amplified band, which simplifies pump management.

SNAIL based TWPA[17] III

The desired three-wave mixing process is

$$\omega_p = \omega_s + \omega_i.$$

This creates signal-idler pair amplification.
However, the same cubic nonlinearity can also generate higher-frequency modes:

$$2\omega_p, \qquad \omega_p + \omega_s, \qquad \omega_p + \omega_i.$$

These modes can take energy away from the desired amplification process and degrade gain.
Fadavi Roudsari *et al.* [17] use periodic variation of circuit parameters to create a stop band near the pump second harmonic:

$$2\omega_p \in \text{stop band}.$$

This suppresses unwanted harmonic and up-converted modes.

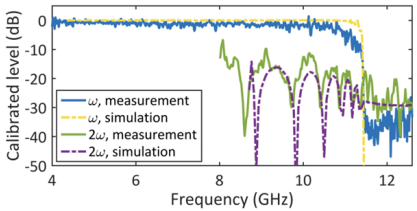


Figure 12: Engineered stop band and harmonic suppression [17].

SNAIL based TWPA[17] IV

The demonstrated SNAIL TWPA achieves $G_{\max} \sim 20\,\mathrm{dB}$, with a 3-dB bandwidth of about $B_{3\mathrm{dB}} \sim 1\,\mathrm{GHz}$, and added noise below one photon. Dispersion engineering does not simply increase the signal gain everywhere. Instead, it performs mode management:

$$\text{desired } \omega_s, \omega_i \quad \text{remain amplified,}$$

while unwanted high-frequency modes are suppressed:

$$\omega_p + \omega_s, \quad \omega_p + \omega_i, \quad 2\omega_p.$$

Tradeoff

The gain profile is not perfectly flat. The usable gain window depends on pump frequency, flux bias, stop-band placement, and the engineered dispersion.

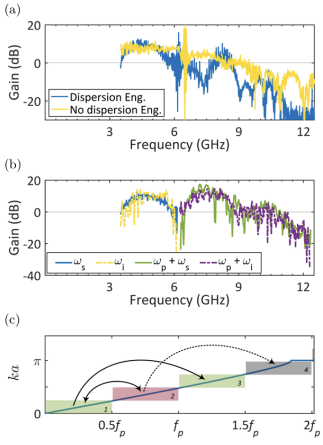


Figure 13: Gain profile and pump-frequency dependence [17].

SNAIL based TWPA[17] V

Advantages and limitations of 3WM SNAIL TWPA

- **Advantage: pump separation.** For 3WM, $\omega_p = \omega_s + \omega_i$, so the pump is far from the amplified signal/idler band, simplifying pump filtering compared with single-pump 4WM.
- **Gain flatness and bandwidth.** The demonstrated SNAIL TWPA reaches up to $20\,\mathrm{dB}$ gain with about $1\,\mathrm{GHz}$ 3-dB bandwidth, but the gain varies noticeably across the band.
- **Unwanted harmonic generation.** 3WM can generate higher harmonics and up-converted tones. Dispersion engineering or stop bands are needed to suppress these unwanted processes.
- **Design sensitivity.** High gain still requires careful phase matching, impedance matching, cutoff-frequency design, and control of fabrication parameter spread.

KI-TWPA[19, 24–28] I

Another realization of nonlinear transmission-line amplification uses the nonlinear kinetic inductance of a superconducting thin film. At large current, the kinetic inductance becomes current dependent and can be phenomenologically written as [19]

$$L_k(I) \approx L_k(0)\left[1+\left(\frac{I}{I_*}\right)^2+\cdots\right], \tag{41}$$

where I_* is a characteristic current scale and $L_k(0)$ is the kinetic inductance in the low-power linear limit. Using $\Phi(t) = L_k(I)I(t)$, we obtain

$$\Phi(t) = L_k(0)I(t) + L_k(0)\frac{I^3(t)}{I_*^2}+\cdots. \tag{42}$$

Thus, the nonlinear term $L_k(0)\frac{I^3(t)}{I_*^2}$ acts like a $\chi^{(3)}$ nonlinearity in optical fibers and naturally generates four-wave mixing.

KI-TWPA[19, 24–28] II

A different operation mode is obtained by applying a dc bias current: $I(t) = I_{\mathrm{dc}} + I_{\mathrm{rf}}(t)$ [19]. Then

$$I^3(t) = (I_{\mathrm{dc}} + I_{\mathrm{rf}})^3 \tag{43}$$

$$= I_{\mathrm{dc}}^3 + 3I_{\mathrm{dc}}^2 I_{\mathrm{rf}} + 3I_{\mathrm{dc}} I_{\mathrm{rf}}^2 + I_{\mathrm{rf}}^3. \tag{44}$$

The term $3I_{\mathrm{dc}}^2 I_{\mathrm{rf}}$ modifies the effective linear inductance, while the term $3I_{\mathrm{dc}} I_{\mathrm{rf}}^2$ acts as an effective second-order nonlinearity for the RF field. Its strength is proportional to the dc bias current I_{dc}.

Therefore, a dc-biased kinetic-inductance transmission line can realize three-wave mixing.

In short, the unbiased kinetic-inductance nonlinearity behaves like a $\chi^{(3)}$ medium and naturally gives four-wave mixing, while a dc bias opens an effective $\chi^{(2)}$-like three-wave mixing process.

Similar to JTWPAs, KI-TWPAs also require dispersion engineering for phase matching. This can be done by engineering the transmission-line impedance, for example by periodically changing L' and C' through CPW/microstrip geometry or artificial loading [20–23]. It can also be done by periodically adding resonant phase shifters along the line [21]. Thus, the phase-matching strategy is similar; what changes is the nonlinear medium.

Advantages and Limitations I

Advantages

- Large dynamic range: the nonlinear interaction is distributed over a long transmission line, which can provide higher power handling.
- **Flexible operating environments:** KI-TWPAs may be useful for $4\,\mathrm{K}$ operation, magnetic-field-compatible readout, and detector-array applications.

Limitations

- **Material loss and film uniformity:** performance depends strongly on the microwave loss, disorder, thickness, and uniformity of high-L_k films.
- **Pump and dc-bias handling:** 3WM operation often requires dc bias and careful pump routing, filtering, and thermalization.
- **Thermal and excess noise:** especially at higher-temperature stages, noise can be limited by thermal population, insertion loss, and excess device noise.

Kerr-Reversal Phase Matching [29, 30] I

Recent Developments I

From mK amplifier to 4 K operation

Malnou *et al.* studied the performance of a kinetic-inductance TWPA at $4\,\mathrm{K}$ [25]. This direction is different from the usual mK-stage quantum-limited preamplifier picture. Instead, the goal is to explore whether KI-TWPAs can serve as low-power cryogenic microwave amplifiers beyond the mixing chamber stage.

This is enabled by the relatively large dynamic range and distributed nonlinearity of kinetic-inductance transmission lines. However, at $4\,\mathrm{K}$, the system noise is no longer limited only by the quantum limit, but also by thermal population, insertion loss, and excess noise.

Recent Developments II

Amplifier and converter

Malnou *et al.* later demonstrated a travelling-wave parametric amplifier and converter [26]. The key idea is that a nonlinear traveling-wave circuit can provide not only forward gain, but also frequency conversion / isolation-like behavior.

This is important because scalable readout chains are limited not only by amplifier performance, but also by bulky nonreciprocal components such as circulators and isolators.

On-chip RF components

Howe *et al.* developed compact superconducting KI-TWPAs with on-chip RF components [27]. This addresses a practical issue: a KI-TWPA requires surrounding microwave infrastructure, such as bias tees, directional couplers, pump routing, and dc-bias networks.

Integrating these components on chip can reduce external microwave hardware, footprint, insertion loss, reflections, and cryogenic wiring complexity.

Future Directions of TWPA

- **Lower added noise:** reduce dielectric loss, insertion loss, pump leakage, and unwanted sideband generation to approach the quantum limit at the system level.
- **Higher saturation power:** support large-scale multiplexed readout by suppressing pump depletion, intermodulation, harmonic generation, and higher-order nonlinear processes.
- **Better dispersion engineering:** control both $k(\omega)$ and Δk_{NL} to obtain flatter gain and reproducible phase matching.
- **Improved pumping schemes:** use 3WM or flux-pumped designs to separate the pump from the signal band, while suppressing harmonic and sum-frequency processes.
- **Nonreciprocal amplification:** combine forward gain with backward isolation or frequency conversion to reduce the need for bulky cryogenic circulators.
- **Broader operating conditions:** extend TWPAs to sub-GHz readout, high-frequency sensing, magnetic-field-compatible environments, and possibly $4\,\mathrm{K}$ operation.
- **Beyond amplification:** use TWPAs as broadband nonlinear microwave platforms for multi-mode squeezing, entanglement generation, and traveling microwave single-photon detection.

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